

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Sentiment Analysis and Opinion Mining

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled “**Sentiment Analysis for Opinion Mining**” in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering (Computer Science and Engineering)**, submitted in the Department of Computer Science and Engineering at **Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India**, is an authentic record of our own work carried out under the supervision of **Dr. Preeti Saini**.

The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

Sentiment analysis has evolved from basic keyword-based approaches to advanced deep learning techniques capable of understanding complex human emotions. Traditional systems fail to capture context, sarcasm, and urgency, leading to inaccurate results.

This project explores the effectiveness of transformer-based models, specifically BERT, for fine-grained emotion detection. Using the GoEmotions dataset, the system classifies text into 27 emotion categories. A comparative analysis with SVM and VADER is conducted based on accuracy, latency, and scalability.

The results demonstrate that BERT significantly outperforms traditional approaches, making it suitable for modern large-scale opinion mining applications.

1. Introduction

- **Problem Background:**

Traditional sentiment analysis systems rely on polarity-based classification (positive/negative), which leads to several limitations:

- **Contextual Blindness:** Misinterpretation of sarcasm and context
- **Computational Inefficiency:** Poor handling of long texts
- **Cascading Inaccuracy:** Missing critical signals like urgency

Additionally, existing research lacks a **fair comparison of BERT, SVM, and VADER under identical conditions**, creating a gap in practical evaluation.

- **Objective:**

- To evaluate whether BERT can replace traditional sentiment analysis models
- To perform **fine-grained emotion classification** across 27 categories
- To compare **accuracy, latency, and scalability** of BERT, SVM, and VADER
- To provide **practical model selection guidance** for different applications

2. Methodology : -

The project is implemented in three phases:

- **Data Preparation:**

Dataset split into training, validation, and testing sets (70/15/15) with preprocessing steps such as lowercasing, emoji conversion, and tokenization

- **Model Training:**

BERT fine-tuned using AdamW optimizer; SVM trained using TF-IDF features

- **Evaluation:**

Performance measured using F1-score, latency, and scalability across multiple nodes

3. Tools and Technologies : -

- **Deep Learning:** PyTorch, HuggingFace Transformers (BERT)
- **Machine Learning:** Scikit-learn (SVM, TF-IDF)
- **Backend:** Java (Spring Boot)
- **Database:** PostgreSQL
- **Infrastructure:** Ubuntu virtual machines, Google Colab GPU
- **Monitoring:** Prometheus, Grafana

4. Implementation : -

- **BERT Model Processing**

BERT processes text using bidirectional context to understand meaning more accurately. This helps capture relationships between words and improves handling of complex language.

- **Fine-Tuning on Dataset**

The model is fine-tuned on the GoEmotions dataset for classification into 27 emotion categories. This allows the model to learn emotion-specific patterns and improve prediction accuracy.

- **Baseline Model Implementation**

SVM (with TF-IDF) and VADER are implemented for performance comparison. These models provide a benchmark to evaluate the effectiveness of the BERT model.

- **Pipeline Architecture**

The system includes preprocessing, model inference, and result storage in a structured pipeline. This ensures smooth data flow from input text to final emotion output.

- **Scalability Testing**

The model is tested in a distributed setup to evaluate performance and scalability. Multiple nodes are used to measure how throughput improves with increased resources.

5. Results : -

- **BERT:**

Accuracy: 92%

F1 Score: 0.88

Best performance in sarcasm detection

- **SVM:**

Accuracy: 78%

F1 Score: 0.76

- **VADER:**

Accuracy: 71%

F1 Score: 0.71

- **Latency:**

VADER: 3ms

SVM: 12ms

BERT: 18ms

- **Scalability:**

BERT achieved near-linear scaling (factor 4.86)

6. Conclusion And Future Scope : -

- **Conclusion:**

BERT significantly outperforms traditional sentiment analysis methods in both accuracy and scalability. It provides deeper emotional understanding compared to polarity-based systems.

- **Future Scope:**

- Hybrid system combining BERT and VADER for optimized performance
- Multilingual support including Hinglish
- Real-time deployment using streaming data
- Enhanced security for sensitive applications

7. References : -

- GoEmotions Dataset (Google Research) : <https://www.kaggle.com/datasets/debarshichanda/goemotions/data>
- Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”
- Hutto & Gilbert, “VADER: A Parsimonious Rule-based Model for Sentiment Analysis”
- Scikit-learn Documentation
- HuggingFace Transformers Documentation